Status	Finished
Started	Sunday, 25 October 2026, 10:58 AM
Completed	Sunday, 25 October 2026, 11:04 AM
Duration	5 mins 38 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection

b. Distribution

c. Extraction

d. Insertion

The correct answer is:
Insertion

Question **2**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470

b. 200,470,80,150,40,90,300,400,70,120

c. 80,150,200,470,40,90,300,400,70,120

d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Dividing the input into smaller subproblems
c. Combining sorted sublists
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It divides the list into two halves, sorts each half, and then merges them
b. It compares adjacent elements and swaps them if necessary
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **5**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Being the most efficient sorting algorithm
c. Sorting data in a non-sequential manner
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **6**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. arrange()
- c. sort()
- d. order()

The correct answer is: sorted()

Question **7**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is less efficient than bubble sort
- b. It does not require recursion
- c. It is based on the divide-and-conquer approach
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **8**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It complicates the analysis process
- c. It allows for quicker identification of duplicates
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **9**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Searching
b. Duplicates
c. Selection
d. All of the above

The correct answer is: All of the above

Question **10**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys
b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
c. A sorting algorithm is stable if it preserves the order of duplicate keys
d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Merge Sort
b. Insertion Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **12**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Bubble sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **13**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass

b. Dividing the input into parts, solving each part, and combining the solutions

c. Rearranging data without sorting

d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **14**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By breaking the input into smaller parts and merging them

b. By using the bubble-up method

c. By comparing elements sequentially

d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **15**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It slows down the process
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items